

Assistive Robotics

A thick, hand-drawn style orange line that spans the width of the title text below it.

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This Lecture



- How should you problem solve for open-ended questions?
- What are the algorithmic approaches to assistive robotics?
- How do we test our hypotheses?

Approaching Problems

To solve open-ended problems either in research or in industry, you need to master two skills: **Vectoring** and **Velocity**.

Vectoring: identify the biggest dimension of risk in your project right now

Velocity: rapidly reducing risk in the chosen dimension

Approaching Problems

To solve open-ended problems either in research or in industry, you need to master two skills: **Vectoring** and **Velocity**.

Explaining Vectoring.

What you *should not* do: “I think I know how to solve this entire problem. I am just going to do that, and try to make it work no matter what.”

What you *should* do: Pick an important aspect of the project, and figure out how to reduce your uncertainty about that question.

Approaching Problems

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Explaining Vectoring.

- Instead of trying to write a complete proof, try to prove the simplest case
- Implement a basic version of your code and test it on a toy problem
- Gather some preliminary data and see if it matches your intuition

Approaching Problems

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Explaining Vectoring.

If you do it right, vectoring makes your research process a loop. You are iteratively identifying the challenging aspects of your system, and building out from there --- likely changing things along the way!

Approaching Problems

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Explaining Velocity.

If you work on a tough problem, you risk getting stuck for extended lengths of time.

- Model is not working well or will not train correctly
- Design not having the intended effect
- Engineering challenges keep coming up

Approaching Problems

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Explaining Velocity.

To be a great engineer, you need to be creative. Velocity is about how rapidly you can explore ways to get unstuck --- try some creative solutions, and see what you learn!

Approaching Problems

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Explaining Velocity.

When you are choosing these new, creative approaches, the thing to remember is your core goal. What is the fundamental idea you want to explore, or fundamental problem you want to solve? Strip everything else away and focus your tests on that core.

Approaching Problems

To solve open-ended problems either in research or in industry, you need to master two skills: **Vectoring** and **Velocity**.

Vectoring: identify the biggest dimension of risk in your project right now
(each week, determine what is the key question you want to solve)

Velocity: rapidly reducing risk in the chosen dimension
(break the question down to its basics, and try several approaches)



Let's apply these
ideas to assistive
robotics



Exoskeletons

There is a trend towards using **softer materials** to reduce weight and enable more flexible motion. The key algorithmic questions focus on:

- What task to assist for (e.g., walking, turning, sitting down)
- How to maintain stability if the human and robot are both moving
- How to ensure energy efficiency for long-term usage
- Personalizing the exoskeleton to the current user

A person is standing and wearing a lower-body exoskeleton. The exoskeleton consists of a black vest-like structure with straps, and leg attachments that extend down to the ankles. The person is also wearing a blue face mask and a head-mounted display. Another person, wearing a black t-shirt, black leggings, and a black face mask, is standing to the left, adjusting the exoskeleton on the person's back. The background is a plain white wall, and the floor is dark. A semi-transparent grey banner is overlaid across the middle of the image, containing the text "Lower-body Exoskeleton Atalante" in white serif font.

Lower-body Exoskeleton Atalante

Exoskeletons

Let's focus on *personalizing* an exoskeleton to the user.

Start by **vectoring**: identify the key question you want to answer first

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Then try **velocity**: what are the first tests you'd want to do to learn about your key question

Exoskeletons

Let's focus on *personalizing* an exoskeleton to the user.

Here is one state-of-the-art approach:

- Given a set of parameters to tune θ
- (If possible) given a prior over those parameters $P(\theta)$
- For each iteration:
 - Sample two different parameter sets θ_1 and θ_2
 - Have the human try walking with each parameter set
 - Ask the human which they preferred (can also have a neutral option)
 - Update your posterior $P(\theta \mid \text{user preference})$

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This method **does not scale** to high-DoF parameter spaces, so we need to identify synergies to reduce the effective dimension of θ

Assistive Robot Arms

The promise of **assistive robot arms** is that they can help humans to everyday tasks (e.g., getting dressed, eating) without relying on a caregiver.

Some of the key algorithmic questions are:

- What does the human want their robot arm to do?
- How do we balance between human and robot control?
- How do we make sure the system is safe?
- How can we make the robot's policy robust to real-world scenarios?

Today's assistive robot arms use **standard utensils** (forks, spoons) to feed adults with motor impairments



[1] "Meet Obi - Adaptive Eating Device [Assistive Technology]"
<https://www.youtube.com/watch?v=Or59UXIRaw&t=3s>



[2] "Autonomous Robot Feeding with Assistive Dexterous Arm (ADA)," <https://www.youtube.com/watch?v=t2eO4CD-0WY>

Assistive Robot Arms

Let's focus on *robot-assisted feeding*. How do we make it robust to real-world environments?

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Assistive Robot Arms

Let's focus on *robot-assisted feeding*. How do we make it robust to real-world environments?

Here is one state-of-the-art approach:

- Given a set of plates, utensils, and chopped food
- For each iteration:
 - Query the human for which item to grasp
 - Attempt to pick up the item
 - If grasp failed, rely on shared autonomy / user feedback to modify the motion
 - Show robot's plan and next steps on a GUI



A big component
of solving these
problems is user
studies

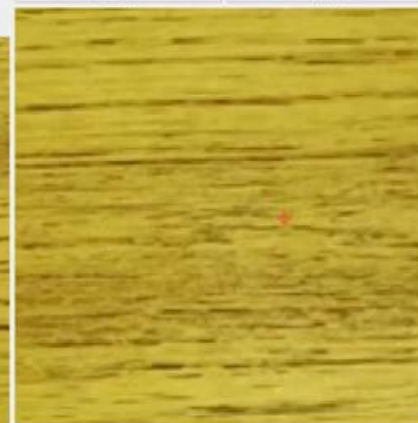




Minimap

Zoom +

Zoom -



Working memory size

Unknown

Long-term memory size

Unknown

GPU memory (all processes, w/ caching)

1.3 GB / 10.8 GB

GPU memory used by torch, w/o caching

668 MB / 10.8 GB

Clear memory

Min. working memory frames

5

Max. working memory frames

10

Max. long-term memory size

10000

Number of prototypes

128

Memory frame every (r)

10

Initialized.

0 / 4369

Play Video

☐ Scribble ☒ Click ☐ Free Brush size: 3

Reset Frame

Overlay Mode davis

Control

Forward Propagate

Backward Propagate

Experimental Design

Independent Variables.

What you control. *Example: the robot's learning algorithm.*

Make sure you **include baselines**

Experimental Design

Independent Variables.

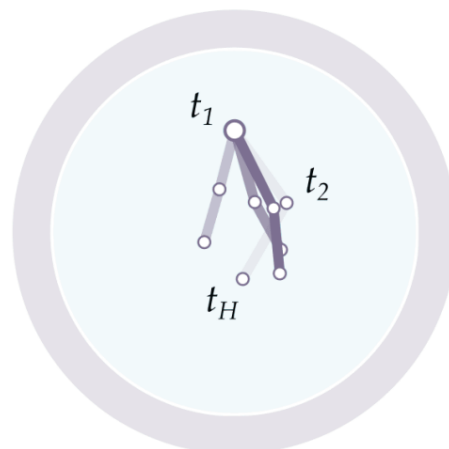
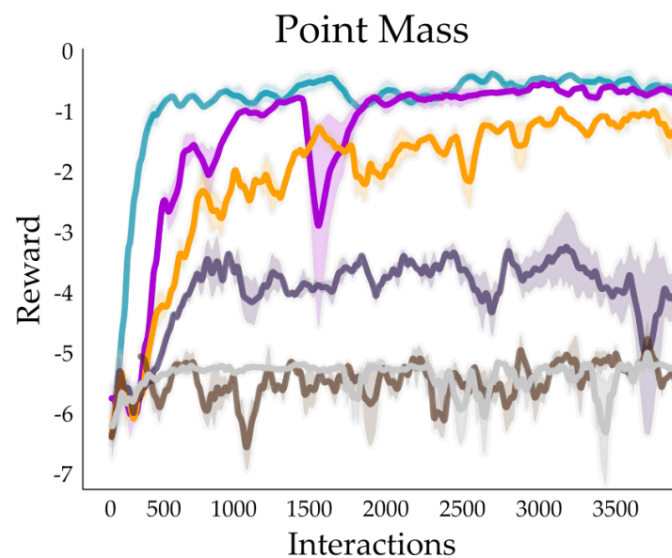
What you control. *Example: the robot's learning algorithm.*

Make sure you include baselines

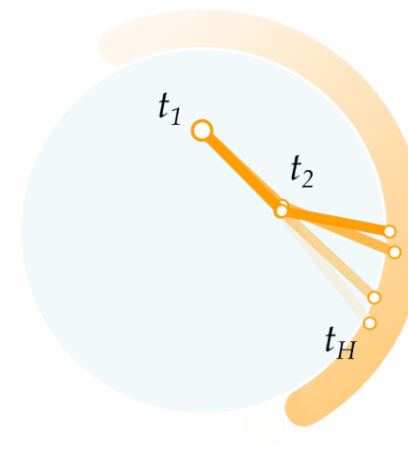
Ablation.

Studies performance of overall approach by removing components.

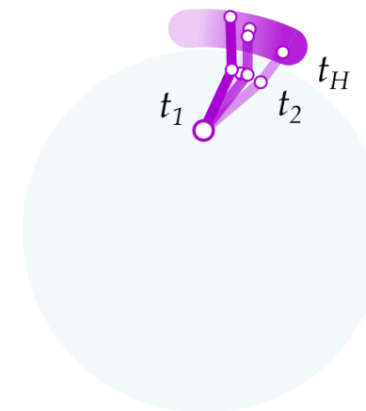
Example: If your approach predicts and plans, then test with just prediction, test with just planning, and then test with both.



LILAC



LILI (no influence)



LILI (ours)

Experimental Design

Dependent Variables.

What you measure. *Example: the time taken to complete a task.*

Often consists of **objective** and **subjective** metrics

*Objective: time, distance,
number of corrections*

Subjective: survey responses

Likert Scale Survey

I knew the robot was not going to collide with me. *

1 2 3 4 5 6 7

strongly disagree ○ ○ ○ ○ ○ ○ ○ strongly agree

I had to move carefully to avoid collisions with the robot. *

1 2 3 4 5 6 7

strongly disagree ○ ○ ○ ○ ○ ○ ○ strongly agree

Likert Scale Survey

1. Flip negative items (8 – answer) so that higher scores are always positive
2. Check **reliability** using Cronbach's α
3. If $\alpha > 0.7$ the scale is reliable
4. Group all items on reliable scale into one **average** score per participant

I knew the robot was not going to collide with me. *

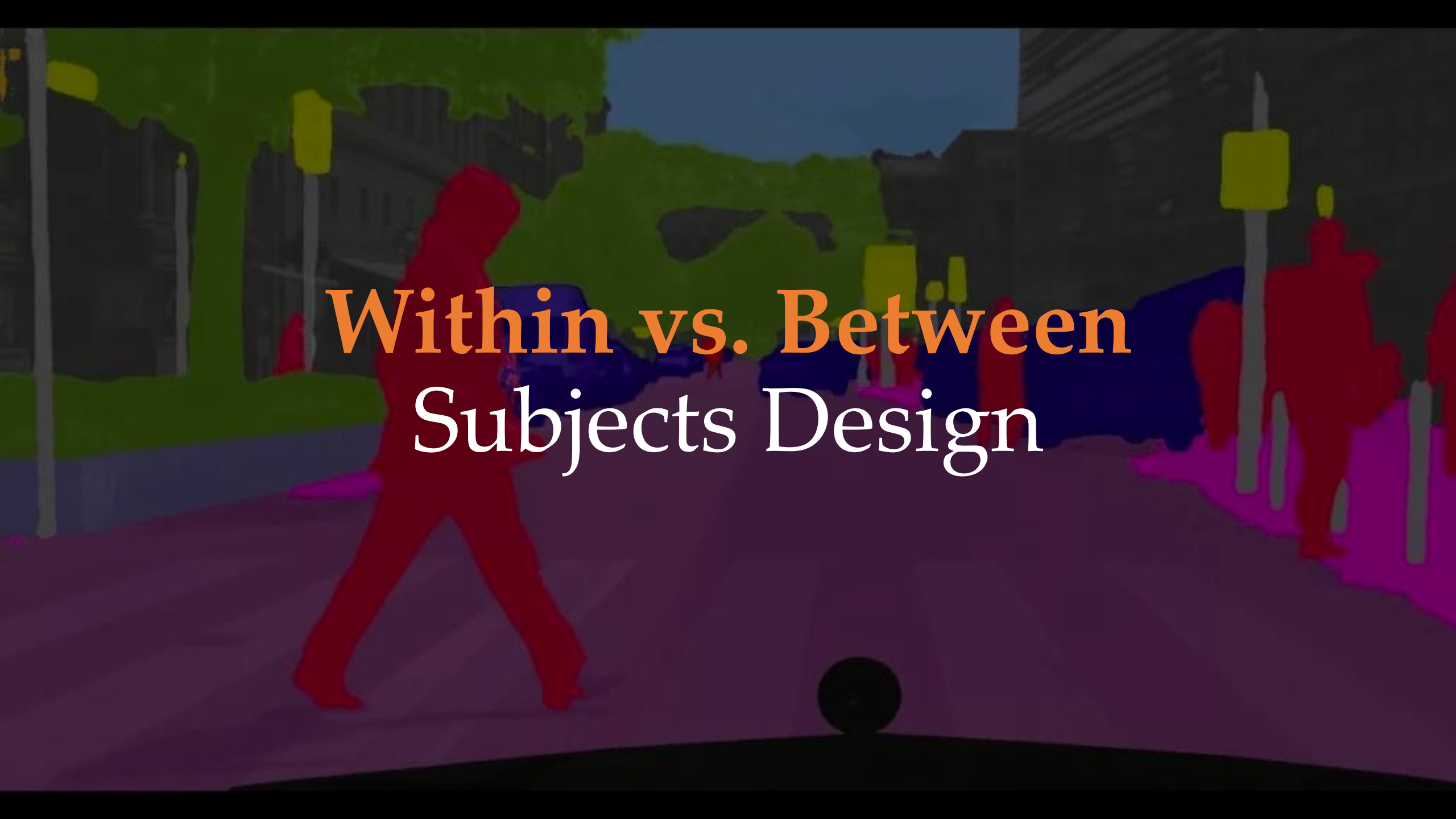
1 2 3 4 5 6 7

strongly disagree ○ ○ ○ ○ ○ ○ ○ strongly agree

I had to move carefully to avoid collisions with the robot. *

1 2 3 4 5 6 7

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Within vs. Between Subjects Design

Procedure

You've recruited X participants for your user study.
Let's say you want to compare two algorithms: A and B.

Between Subjects.

- Half of the participants interact with A, the other half interact with B.

Within Subjects.

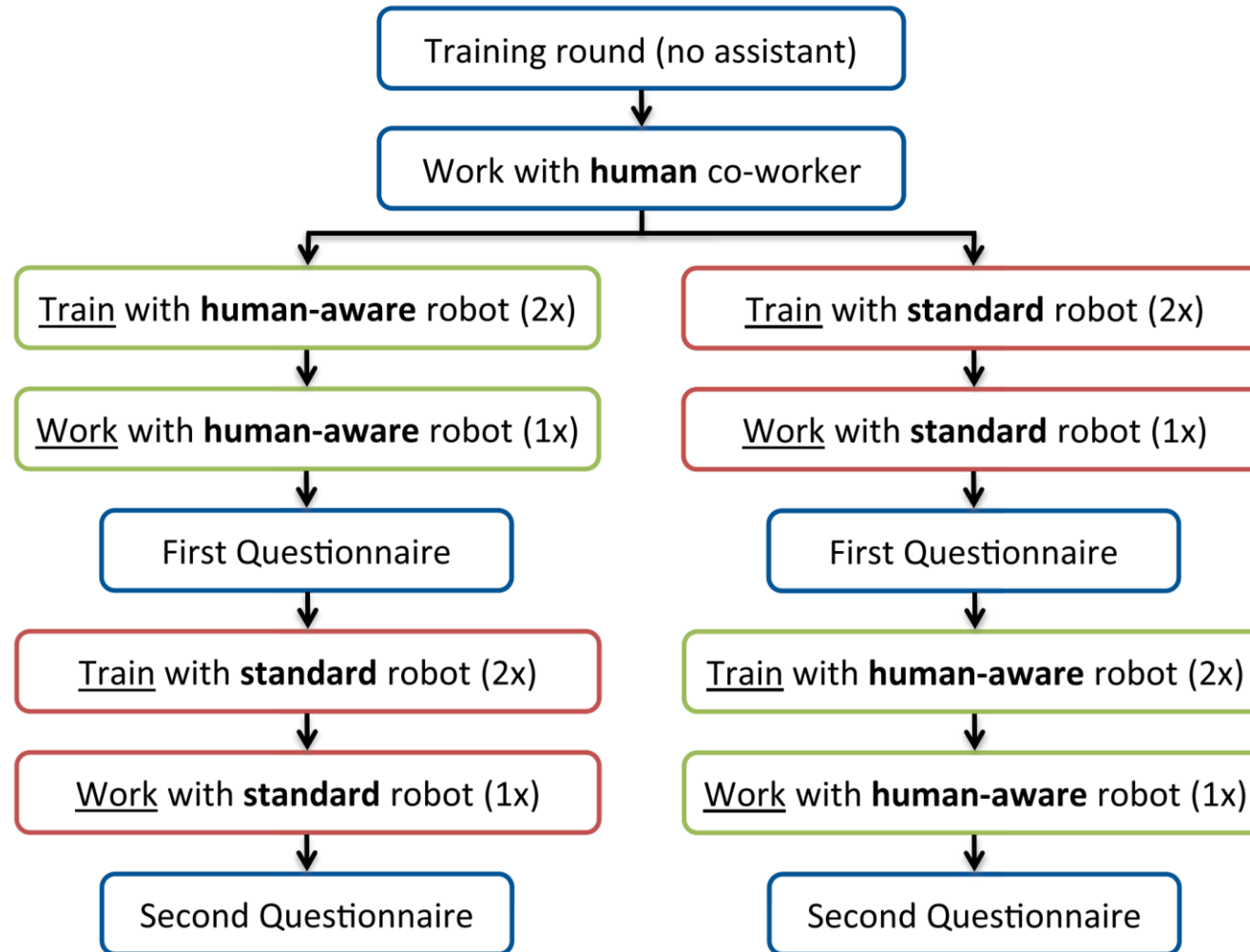
- Every participants works with A and B

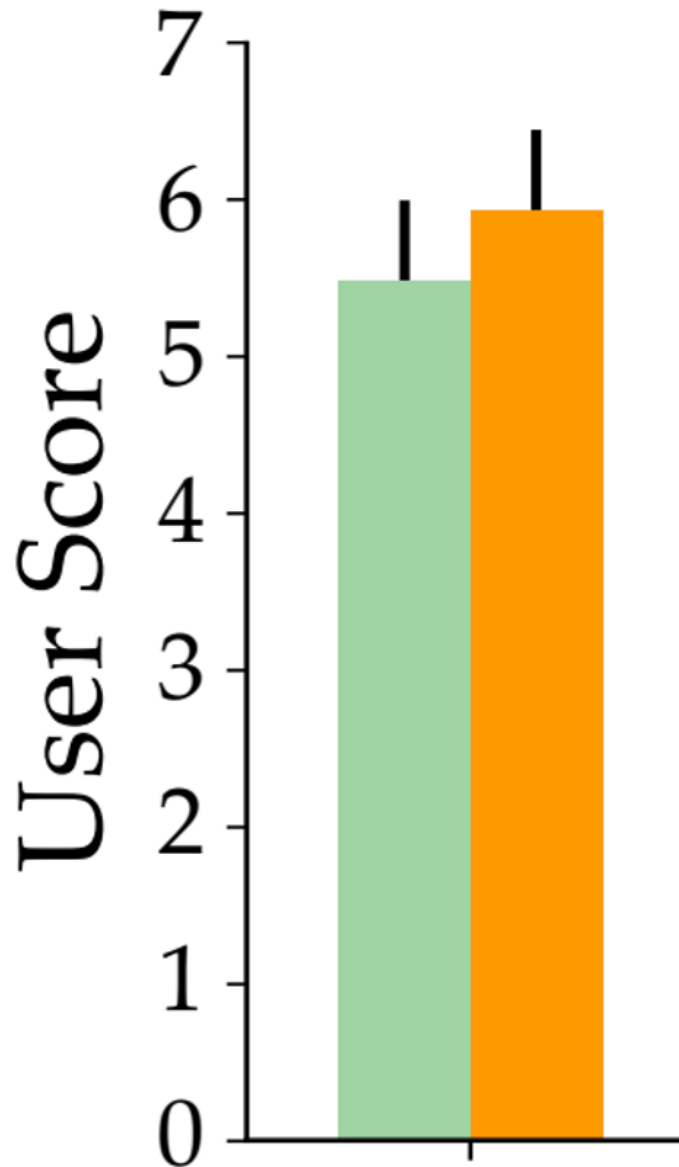
Procedure

You've recruited X **participants** for your user study.
Let's say you want to compare two algorithms: A and B.

Within Subjects.

- Every participants works with A and B
- The order of presentation is important: *if everyone works with A first, they may end up doing better with B because of experience*
- We **counterbalance** the order so half start with A and the others start with B





Motivation

- Our results show that **B** scored higher on average than **A**
- But is this difference just because of noise? Or is this a *significant* difference in performance?

NULL HYPOTHESIS

THE NULL HYPOTHESIS ASSUMES
THERE IS NO DIFFERENCE BETWEEN
TWO GROUPS.

NULL HYPOTHESIS

H_0

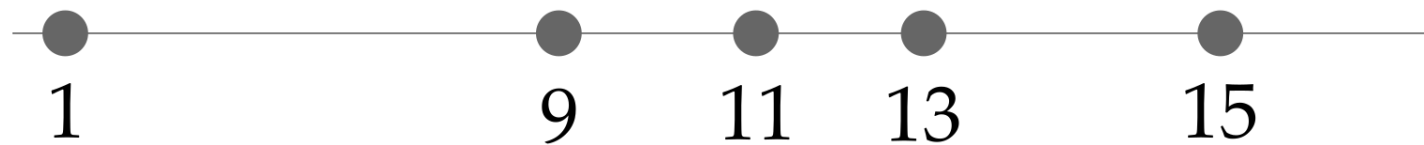
LIGHT COLOR HAS NO EFFECT
ON PLANT GROWTH

ALTERNATIVE HYPOTHESIS

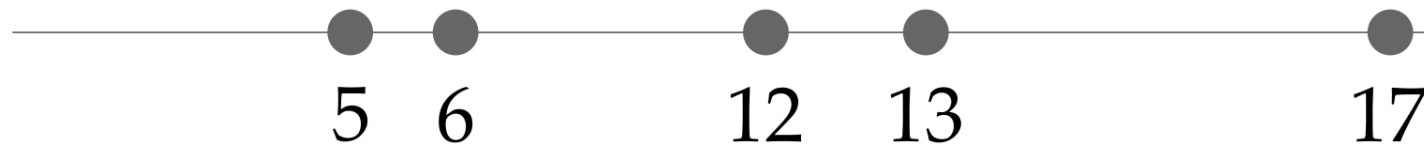
H_A

LIGHT COLOR AFFECTS PLANT
GROWTH

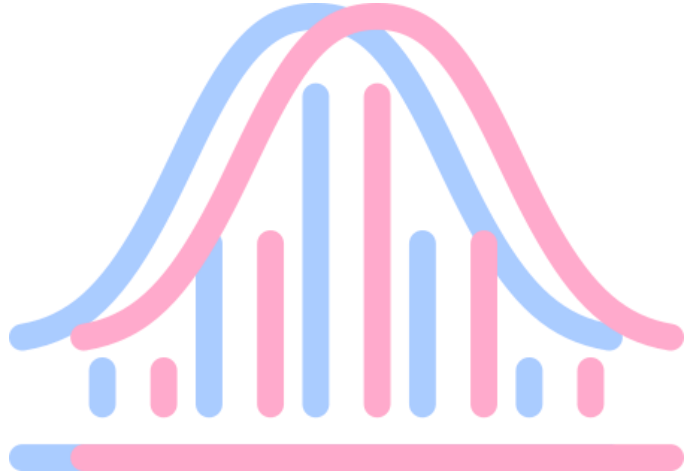




Height of the plants with blue light

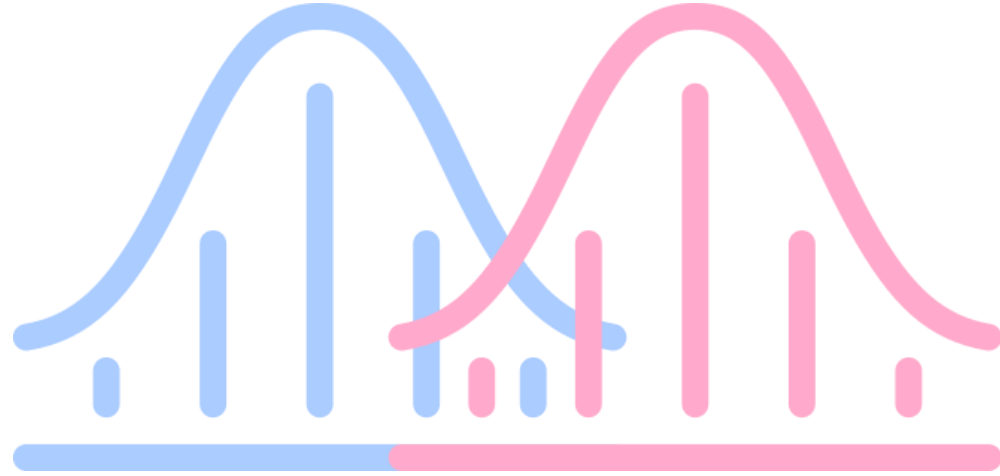


Height of the plants with red light



Null Hypothesis

Heights are close, we don't have enough evidence to say that red is higher than blue.



Rejecting the Null Hypothesis

Heights are different enough that we are pretty confident that red does lead to taller plants than blue

Statistical Significance

You collected results with two algorithms: A and B

- The null hypothesis is that the algorithm type had no effect: differences in average scores are due to random chance
- Rejecting the null hypothesis means we are confident that algorithm type does influence the results

Testing for **statistical significance** is testing whether we can reject the null hypothesis.

A white Pioneer 3-DX mobile robot is positioned in the center of the frame, navigating through a field of young corn plants. The robot has a white body with a black sensor array on top and a black logo on the front. It is surrounded by lush green corn leaves, with sunlight filtering through the foliage, creating a dappled light effect on the ground and the robot. The text "What are some common statistical tests?" is overlaid on the image, with "What are some common" in white and "statistical tests?" in orange.

What are some common
statistical tests?

T-Test

Independent Variables.

You are comparing two groups: A and B

Protocol.

- If using a **between subjects** protocol, use *Independent t-test*
- If using a **within subjects** protocol, use *Dependent t-test*
- Reject the null hypothesis if $p < .05$

ANOVA

Independent Variables.

You are comparing three or more groups: A, B, and C

Protocol.

- If using a **between subjects** protocol, use *one-way ANOVA*
- If using a **within subjects** protocol, use *repeated measures ANOVA*
- Reject the null hypothesis if $p < .05$

This Lecture



- How should you problem solve for open-ended questions?
- What are the algorithmic approaches to assistive robotics?
- How do we test our hypotheses?

Next Lecture



- How can humans and robots communicate?